Kevin Jacob

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Education

Bachelor of Science in Computer Science Dec 2027
Georgia Institute of Technology, Atlanta, GA *GPA: 3.95*

- Concentrations: Artificial Intelligence and Theory
- Courses: Efficient ML (FSDP, Quantization), Machine Learning, Intro to AI, Graph Theory, Automata & Complexity

Experience

Apple May 2026 – Aug 2026
Software Engineering Intern *Cupertino, CA*

- Engineered a Python Slack bot module (Bolt) that ingests CI tooling support requests and pushes data via S3 to Snowflake, automating metric collection and saving 150+ engineering hours annually
- Modeled 30+ operational metrics with dbt SQL transformations on Snowflake, surfacing trends in a Streamlit UI used for CI triage and reviewed quarterly with partner tool teams
- Developed an ML pipeline that strips PII from support requests, clusters via HDBSCAN, and applies LLM-generated labels to each cluster, surfacing 60+ issue categories to guide tooling and documentation priorities

Vantaca + HOAi (YC W22) May 2025 – Dec 2025
Software Engineering Intern *Wilmington, NC*

- Built a LangGraph agent microservice triggered on Jira updates that auto-generates documentation from git PRs and org context via semantic search, accelerating doc workflows by 3× across 50+ tickets/week; monitored in LangSmith
- Deployed agent via dockerized Azure Function with CI/CD pipeline, achieving 95% quality score in tech writer evals
- Architected hosted MCPs via Azure App Services with custom OAuth for secure LLM knowledge access by 200+ staff

Research

Graph Computation and Machine Learning Lab - Georgia Tech Nov 2025 – Present
ML Research Intern *Atlanta, GA*

- Co-authored SparseExpansion (under review, AISTATS 2027), a differentially private subgraph sampler that randomly drops subgraph edges to amplify node-level privacy, cutting ε from $\sim 10^3$ to ≤ 10 for multi-hop GNNs
- Ran experiments in PyTorch Geometric and Opacus on H200 GPUs via SLURM, benchmarking against 4 private baselines on 7 graphs (up to 2.4M nodes, 124M edges); retains $\sim$95% of non-private accuracy at $\varepsilon = 8$

Assistive Intelligence Lab - Georgia State University May 2023 – Present
ML Research Intern *Atlanta, GA*

- Proposed NRA-Embed, combining a noise-robust contrastive objective with annotator embeddings to separate genuine disagreement from label noise; matches SOTA on clean data and beats baselines by up to 20 F1 at 40% noise
- Implemented a benchmarking harness in PyTorch running 500+ RoBERTa fine-tuning jobs across 4 datasets, 2 synthetic noise models, and 5 seeds on a Linux-based on-premise cluster (1–12 GPUs); cut training time by 80%

Publications

When Disagreement Meets Noise: Noise Robust Annotator Embeddings for Subjective NLP

- **Jacob, K.**, Ghazvinian, P., Jinadu, U., Yang, N. T., & Ding, Y. | *Findings of EMNLP 2026*

Projects

ChatGT (Team Lead) - GT/ NVIDIA AI Makerspace | *TypeScript, React, Ollama, sqlite-vec* Aug 2025 – Present

- Shipped a fully local LLM study assistant that generates adaptive quizzes via JSON-schema structured outputs
- Upgraded RAG retrieval to dense embeddings in sqlite-vec, lifting top-1 slide accuracy 45%→66% over keyword search

Event Tracking Platform - Alpha Kappa Psi | *Rust, React, MongoDB, AWS Lambda* Dec 2024 – Dec 2025

- Built event tracking app in Rust & React with MongoDB and REST APIs on AWS Lambda, serving 300+ active users
- Led a 9-engineer team through Agile sprints as project manager, shipping real-time check-in and scheduling features

Skills

Programming & Frameworks: Python, Java, C/C++, Rust, Swift, TypeScript, JavaScript, SQL, React
ML & AI: PyTorch, PyTorch Geometric, Opacus, LangGraph, Ollama, RAG, MCP, Scikit-learn, Pandas, OpenCV
Data & Infra: Snowflake, dbt, AWS, Azure, GCP, Docker, SLURM, MongoDB, SQLite, Streamlit, Firebase